

Capital, Ideas, and the Costs of Financial Frictions

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Overview

Incumbent firms choose how much to **invest** and **innovate**.

Not the story of entrants v/s incumbents!

- ▶ How do financial frictions distort this decision?
- ▶ Does it quantitatively matter?

This paper:

- ① **Pecking order of firms:** Low network (size, age) firms invest more, high network firms innovate more.
- ② **Endogenous growth model:** Het. firms + financial frictions.
 - ▶ Return to capital v/s innovation generates pecking order.
 - ▶ Frictions slow capital accumulation and therefore shift to innovation.
- ③ **Quantify output losses:** Lower growth/ innovation reduces GDP by 25% over 50 years, capital misallocation by 5%.

Punchline

Two costs of financial frictions:

- ① ↓ **TFP level** due to *misallocation* (distorted capital, *given* productivity).
- ② ↓ **TFP growth** rate due to *lower innovation by constrained firms*.

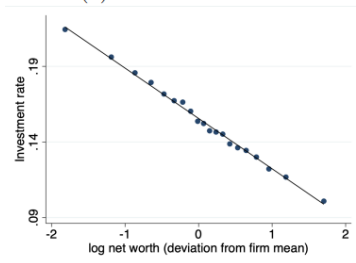
"It's not that good ideas lack funding (capital misallocation), but that fewer good ideas are discovered in the first place (lower innovation)."

Outline

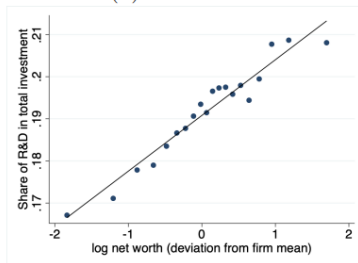
- ① Empirical Facts
- ② Model
- ③ Calibration
- ④ Quantifying Costs

Pecking Order of Firm Growth

(a) Investment rates



(b) R&D share

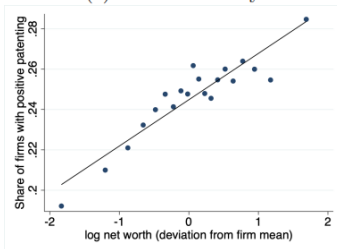


Data: Annual US firm-level Compustat, 1975-2018.

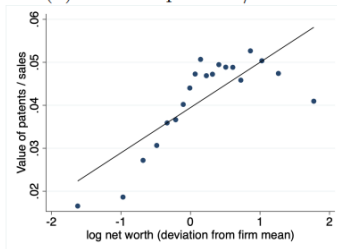
Investment rate _t	Capex _t / Plant, Property, and Equipment (PPE) _{t-1} .
R&D share _t	R&D Exp _t / (R&D Exp _t + Capex _t).
Net worth _t	PPE _t + EBITDA _t + (R&D Exp _t / PPE _{t-1}) + Short-term Investments _t - Total Debt _t .

Pecking Order of Firm Growth

(c) Patent activity



(d) Value of patents / sales



Data: Annual US firm-level Compustat, 1975-2018.

Patents_t	$\# \text{ Patents filed}_t$
$\text{Patent-value-to-sales}_t$	$\text{Patents' mkt value}_t \text{ (Kogan et al. 2017}^1\text{)} / \text{Avg. sales in last 5 yrs.}$
Net worth_t	$\text{PPE}_t + \text{EBITDA}_t + (\text{R\&D Exp}_t / \text{PPE}_{t-1}) + \text{Short-term Investments}_t - \text{Total Debt}_t.$

¹Change in firm equity value in a narrow window around patent approval, in 1982\$, deflated by CPI.

Pecking Order of Firm Growth - Summary

As firms accumulate net worth:

- 1 Investment activity *decreases*.
- 2 Innovation activity *increases*.

Infer: Firms with high shadow price of external finance (low net worth/ small size/ young), face a high return to capital but low return to innovation.

Robustness checks:

- ▶ Other measures of shadow price of external finance: ▶ Size ▶ Age
- ▶ ▶ Other innovation measures.
- ▶ ▶ Orbis data.
- ▶ v/s small firms do more innovation than large firms literature. ▶ Firm FE
- ▶ Other permutations of FE. ▶ Other FE

Outline

- ① Empirical Facts
- ② Model
- ③ Calibration
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Model Overview

Discrete time, no aggregate uncertainty.

- ① **Representative households** with preferences: $\sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\sigma} - 1}{1-\sigma}$. Firms discount using $\frac{1}{1+r_t} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma}$.

② **Heterogeneous firms:**

- Exogenous **exit shock** π_d at beginning of period: Produce, sell capital, pay debt. Replaced by **entrants**: debt = 0, draw from $\Phi_t^0(z, k)$: Imitation for +ve growth of BGP.
- **Continuing firms** choose investment k_{jt+1} , innovation i_{jt} and debt b_{jt+1} .
- Distribution over (z_{jt}, n_{jt}) .

③ **Prices:** Real interest rate r_t obtained from market clearing condition:

$$C_t = \int \left(y_{jt} - (k_{jt+1} - (1 - \delta)k_{jt}) - (A_t z_{jt})^{\frac{1}{1-\alpha}} i_{jt} \right) dj.$$

Continuing Firms

$$v_t^{\text{cont}}(z, n) = \max_{k', i, b'} \underbrace{A_t z k^\alpha + (1 - \delta)k - b - k'}_n \underbrace{- (A_t z)^{\frac{1}{1-\alpha}} i + \frac{b'}{1 + r_t}}_d$$

$$+ \frac{1}{1 + r_t} [\pi_d \mathbb{E}_t(n') + (1 - \pi_d) \mathbb{E}_t(v_{t+1}^{\text{cont}}(z', n'))]$$

s.t. $b' \leq \theta k'$, $d \geq 0$ and $i \geq 0$.

- Undifferentiated good.
- $\alpha < 1 \Rightarrow \exists k_{jt}^*(z_{jt})$.
- $A_t = (\int z_{jt} dj)^a$, $a \geq 0$ is degree of spillovers.
- $(A_t z)^{\frac{1}{1-\alpha}}$: Cost grows to ensure BGP exists.
- π_d : Exit shock so do not outgrow financial frictions in long run.

Continuing Firms

$$v_t^{\text{cont}}(z, n) = \max_{k', i, b'} \underbrace{A_t z k^\alpha + (1 - \delta)k - b - k'}_n \underbrace{- (A_t z)^{\frac{1}{1-\alpha}} i + \frac{b'}{1+r_t}}_d$$

$$+ \frac{1}{1+r_t} [\pi_d \mathbb{E}_t(n') + (1 - \pi_d) \mathbb{E}_t(v_{t+1}^{\text{cont}}(z', n'))]$$

s.t. $b' \leq \theta k'$, $d \geq 0$ and $i \geq 0$.

- $b' \leq \theta k'$: Collateral borrowing constraint. $\mu_t(z, n)$
- $d \geq 0$: No equity issuance. $\lambda_t(z, n)$
- $i \geq 0$: Generate inaction in innovation (data: Zero R&D exp in 50% obs.)
- $\log z_{jt+1} = \begin{cases} \log z_{jt} + \Delta + \varepsilon_{jt+1} & \text{with probability } \eta(i_{jt}) \\ \log z_{jt} + \varepsilon_{jt+1} & \text{with probability } 1 - \eta(i_{jt}) \end{cases}$ ►► Lifecycle

Firm Decision Rules (Proposition 1)

Thresholds $\bar{n}_t(z)$ and $\underline{n}_t(z, n)$ partition the state space: ▶ Eqn

Financially unconstrained

$$n \geq \bar{n}_t(z)$$

$$d > 0, \quad b' < \theta k'$$

Currently constrained

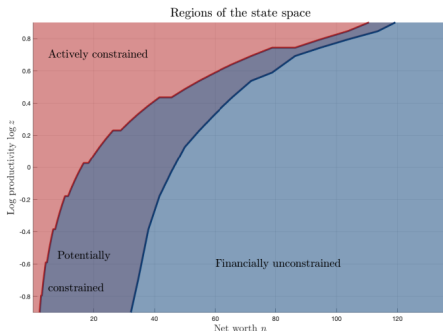
$$n \leq n_+(z, n)$$

$$d = 0, \quad b' = \theta k'$$

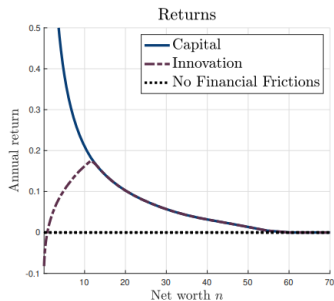
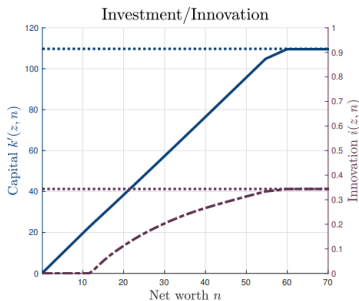
Potentially constrained

$$n \in (\underline{n}_t(z, n), \bar{n}_t(z))$$

$$d = 0, \quad b' < \theta k'$$



Pecking Order of Firm Growth in the Model



» $\eta(i)$

» EBCs

» Het. Δ

» Across z

» FOCs

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Calibration

FITTED PARAMETERS AND EMPIRICAL TARGETS

Parameter	Description	Value	Target (all joint)	Data	Model
<i>Innovation technology</i>					
η_0	Idea arrival function	1.01	Regression coefficient	1.09	1.03
η_1	Idea arrival function	0.21	$\mathbb{E}[\text{RD}_{jt}/y_{jt} \text{RD}_{jt} > 0]$	0.06	0.06
Δ	Size of innovations	0.10	$\mathbb{E}[x_{jt}/k_{jt} \text{spike}]$	0.37	0.35
<i>Financial frictions</i>					
θ	Collateral constraint	0.52	$\mathbb{E}[b_{jt}/k_{jt}]$	0.34	0.30
<i>Productivity shocks</i>					
σ_ε	SD of shocks	0.03	$\sigma(x_{jt}/k_{jt})$	0.15	0.13
<i>Innovation spillovers</i>					
a	Innovation spillovers	0.55	Growth rate	0.02	0.02

Probability of success: $\eta(i) = 1 - \exp\{-\eta_0 [(1+i)^{\eta_1} - 1]\}$ (Increasing, concave, bounded between 0 and 1, and $\eta(0) = 0$).

For **probability** $\eta(i)$ and **size** Δ of a successful innovation, use **investment spikes**², because...

» Others

²> 20% following Cooper and Haltiwanger (2006)

Untargeted Moments

Statistic	Data	Model
<i>Investment spending</i>		
$\mathbb{E}[x_{jt}/k_{jt}]$	0.15	0.16
$\sigma(x_{jt}/k_{jt})$ (targeted)	0.15	0.13
$\mathbb{E}[x_{jt}/k_{jt} \text{spike}]$ (targeted)	0.37	0.35
<i>R&D spending</i>		
$\mathbb{E}[\text{RD}_{jt}/y_{jt}]$	0.06	0.03
$\text{Frac}(\text{RD}_{jt}/y_{jt} > 0)$	0.45	0.92
$\mathbb{E}[\text{RD}_{jt}/y_{jt} \text{RD}_{jt} > 0]$ (targeted)	0.06	0.06
<i>Leverage</i>		
Mean gross leverage, all (targeted)	0.34	0.30
Mean gross leverage, Compustat	0.21	0.28
SD gross leverage, Compustat	0.22	0.24
Mean net leverage, Compustat	0.13	0.09
SD net leverage, Compustat	0.33	0.36

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Costs of Financial Frictions

Two costs of financial frictions:

- ① ↓ **TFP growth** rate g because *lower innovation by constrained firms*:

$$g \approx \frac{1}{1-\alpha} (1+a) (e^{\Delta} - 1) \int l_{jt} dj$$

►► Derivation

- ② ↓ **TFP level** due to *misallocation* (distorted capital allocation across firms with different TFP).
-

- Difficult to decompose because changes in allocation of capital also affect innovation decisions. Upper bound for the misallocation costs:

$$Y_t^* = \max_{k_{jt}} A_t \int z_{jt} k_{jt}^{\alpha} dj \quad \text{such that} \quad \int k_{jt} dj \leq K_t$$

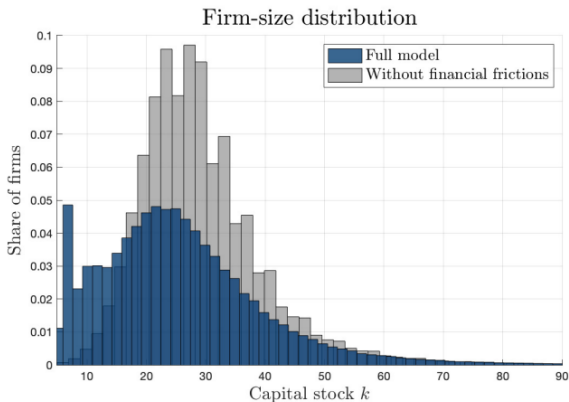
►► Decision

Aggregate Output Losses from Financial Frictions

	Lost growth per year	20 years	30 years	40 years	50 years	Misallocation costs (upper bound)
Baseline	41bps	8.6%	13.1%	17.9%	22.8%	5.0%
Higher η_0	54bps	11.0%	16.9%	23.2%	29.8%	4.8%
Higher η_1	54bps	11.0%	16.9%	23.2%	29.8%	4.6%
Higher θ	33bps	6.7%	10.2%	13.8%	17.5%	3.9%
Lower a	39bps	7.9%	12.1%	16.5%	21.1%	5.2%
Lower z_0	35bps	7.1%	10.8%	14.7%	18.7%	3.8%
Higher k_0	39bps	7.9%	12.1%	16.5%	21.0%	4.6%
↓				↓		↓
$g^{**} - g^m$				$Y^{**} =$		$Y^* =$
				$1.23Y^m$		$1.05Y^m$

- ▶ g^m, Y^m : Calibrated model.
- ▶ g^{**}, Y^{**} : Frictionless model - all firms follow unconstrained policies $k^*(z)$ and $i^*(z)$.
- ▶ Y^* : Frictionless model - but only with undistorted k .

Distributional Effects of Financial Frictions



Distribution of capital along BGP: Detrended to compute stationary distribution.

Planner's Allocation

- **Constrained-efficient planner:** Faces same constraints, but internalizes positive spillovers from innovation.

- Augmented Bellman equation:

$$\omega_t^{\text{cont}}(z, n) =$$

$$\max_{k', b', i} n - k' - (A_t z)^{\frac{1}{1-\alpha}} i + \frac{b'}{1+r_t} + \Lambda_t z + \frac{1}{1+r_t} \mathbb{E}_t [\omega_{t+1}(z', n')] \text{ s.t.} \\ d \geq 0 \text{ and } b' \leq \theta k'$$

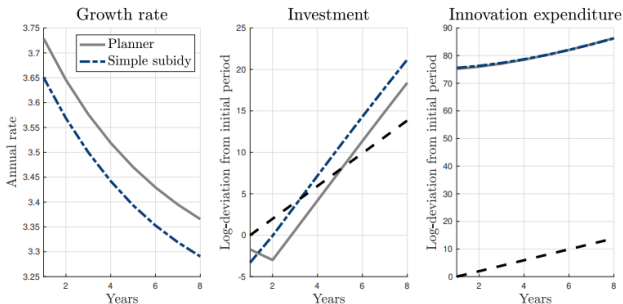
- Shadow value of innovation externality:

$$\Lambda_t = \underbrace{a \left(\int z_{jt} dj \right)^{a-1}}_{\text{Marginal } \uparrow \text{ Agg. Prod.}} \times \underbrace{\int (1 + \lambda_{jt}) \left[z_{jt} k_{jt}^\alpha - \frac{1}{1-\alpha} \left(A_t^{\frac{\alpha}{1-\alpha}} z_{jt} \right)^{\frac{1}{1-\alpha}} i_{jt} \right] dj}_{\text{Marginal } \uparrow \text{ cashflows}}$$

- Financial frictions $(1 + \lambda_{jt})$ amplify positive externality.

Planner's Allocation

- Prefers **more innovation** but **tradeoff** for investment: ► Decisions
- **Constrained:** Higher innovation $\Rightarrow$ less investment (substitutes).
 - **Unconstrained:** Higher innovation $\Rightarrow$ more investment (complements).



Transition Path: Planner starting from market equilibrium BGP.

Conclusion and Comments

- Literature: Frictions distort investment decisions and misallocate capital.
- This paper: Frictions distort innovation and lower economic growth.

"It's not that good ideas lack funding (capital misallocation), but that fewer good ideas are discovered in the first place (lower innovation)."

- ▶ Very small sample of firms! About 30,000 firms whereas there are over 6 million firms in the US.
- ▶ Nature of capital may be different after successful innovation. Existing v/s new line of operation.
- ▶ Δ calibration a bit fishy: Size of investment spike which is short-lived in data but that higher delta is forever in model?

APPENDIX

Sample: Innovative Firms in Incumbent Phase

Sample **excludes**:

- Firms in finance, insurance, real estate, utilities³, nonoperating establishments⁴, industrial conglomerates.
- Firms not incorporated in the United States.
- Any of the following firm-year observations:
 - Negative assets, sales, capital expenditure, or R&D.
 - Low capital values (gross PPE⁵ below \$5M in 1990 dollars).
 - Acquisitions larger than 20% of assets.
 - Investment rates higher than 1.
 - R&D-to-sales ratios higher than 0.3.
 - Gross leverage (total debt/total assets) higher than 10 or negative.
- Firms with less than 20 years of data.

» Pecking Order

³Office of Energy & Transportation

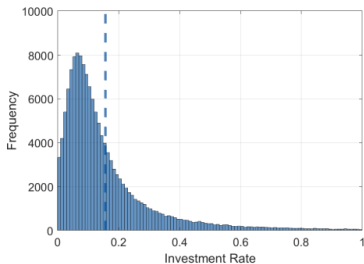
⁴Do not own, lease any tangible property material to the Business

⁵Plant, Property, Equipment

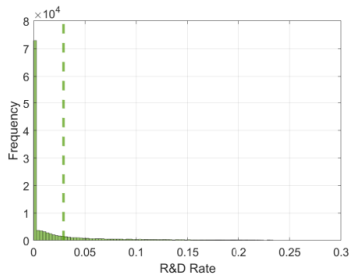
Descriptive Statistics 1

FIGURE A.1: Distribution of Investment Rates and R&D

(a) Investment rates



(b) R&D-to-sales



Notes: This figure shows the histogram of investment rates and the R&D-to-sales ratio. Vertical dashed lines represent each variable mean. For variables definitions and sample selection, see Appendix A.1.

►► Pecking Order

Pecking Order - By Size

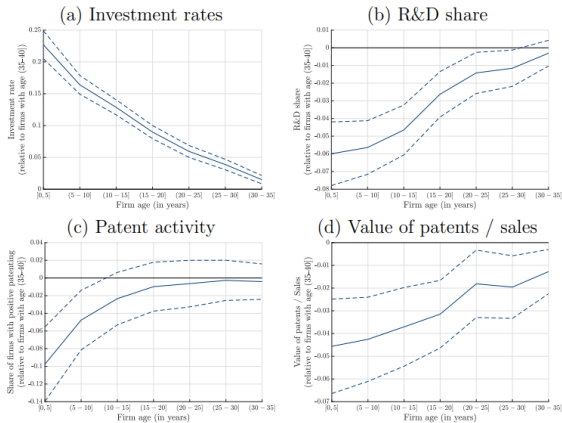
THE PECKING ORDER BY VARIOUS MEASURES OF SIZE

	(1) Investment rate	(2) R&D share	(3) Patent activity	(4) Patent-value -to-sales
log net worth				
$\hat{\gamma}$	-0.068 (0.003)	0.024 (0.003)	0.049 (0.007)	0.021 (0.005)
N	45935	47286	49105	31176
R^2	0.263	0.857	0.639	0.678
log capital				
$\hat{\gamma}$	-0.087 (0.003)	0.026 (0.003)	0.064 (0.008)	0.022 (0.005)
N	49986	51569	53625	33561
R^2	0.272	0.852	0.633	0.671
log capital including intangibles				
$\hat{\gamma}$	-0.087 (0.003)	0.039 (0.004)	0.046 (0.009)	0.020 (0.005)
N	44471	44794	46484	30549
R^2	0.282	0.852	0.634	0.670
log employment				
$\hat{\gamma}$	-0.045 (0.005)	0.002 (0.005)	0.110 (0.011)	0.008 (0.006)
N	45050	45827	47508	30868
R^2	0.207	0.851	0.636	0.673
log sales				
$\hat{\gamma}$	-0.058 (0.004)	0.018 (0.004)	0.094 (0.010)	0.026 (0.005)
N	49986	51569	53625	33561
R^2	0.218	0.851	0.634	0.671
Mean	0.13	0.16	0.34	0.05

$o_{jt} = \alpha_j + \gamma \log s_{jt} + \epsilon_{jt}$, o_{jt} is outcome of interest, s_{jt} is measure of size. Capital=PPE, Intangibles=R&D.

Pecking Order - By Age

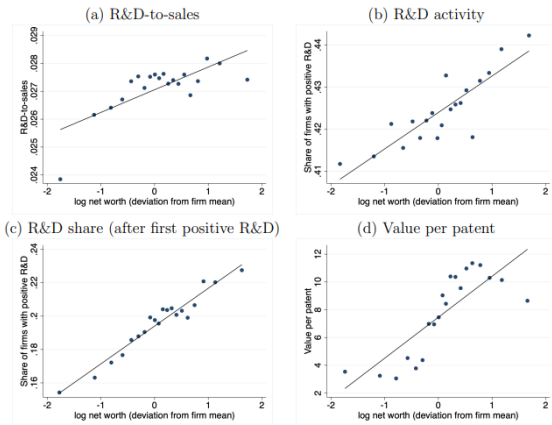
FIGURE 2: The Pecking Order by Age



$o_{jt} = \alpha_j + \sum_{s \in S} \gamma_s \text{ age}_{s jt} + \epsilon_{jt}$, $\text{age}_{s jt}$ is 1 if firm's age in period t is within age bin $s \in S \equiv \{[0, 5], (5, 10], (10, 15], (15, 20], (20, 25], (25, 30], (30, 35]\}$. Datastream (age since incorporation).

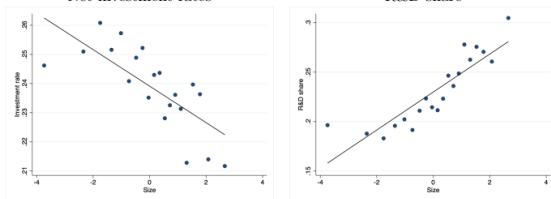
Pecking Order - Other Measures of Innovation

FIGURE A.3: The Pecking Order of Firm Growth for Other Measures of Innovation



Pecking Order - Orbis data

FIGURE A.2: The Pecking Order of Firm Growth in Orbis Data
Net investment rates
R&D share



» Back

Pecking Order - Within Firm

TABLE A.2
ROLE OF WITHIN-FIRM VARIATION FOR VARIOUS MEASURES OF SIZE

	R&D share		R&D-to-sales		Patent activity		Patents per employee		Value of patents / sales	
	(1) No FE	(2) Firm FE	(3) No FE	(4) Firm FE	(5) No FE	(6) Firm FE	(7) No FE	(8) Firm FE	(9) No FE	(10) Firm FE
(a) Unrestricted sample										
log net worth	-0.021 (0.001)	0.021 (0.002)	-0.001 (0.000)	0.002 (0.000)	0.101 (0.001)	0.057 (0.004)	0.210 (0.013)	0.180 (0.061)	0.032 (0.001)	0.016 (0.003)
log capital	-0.051 (0.001)	0.021 (0.002)	-0.006 (0.000)	0.001 (0.001)	0.089 (0.001)	0.066 (0.005)	-0.063 (0.011)	-0.063 (0.072)	0.026 (0.000)	0.016 (0.004)
log employment	-0.032 (0.001)	0.003 (0.003)	-0.005 (0.000)	0.002 (0.001)	0.102 (0.001)	0.084 (0.005)	-0.232 (0.012)	-0.392 (0.076)	0.024 (0.000)	0.008 (0.003)
log sales	-0.032 (0.001)	0.014 (0.002)	-0.005 (0.000)	0.002 (0.001)	0.108 (0.001)	0.080 (0.005)	-0.061 (0.012)	-0.087 (0.077)	0.029 (0.000)	0.020 (0.004)
(b) Firms with more than 20 years of data										
log net worth	-0.011 (0.001)	0.024 (0.003)	0.001 (0.000)	0.002 (0.001)	0.103 (0.002)	0.049 (0.007)	0.256 (0.018)	0.272 (0.096)	0.039 (0.001)	0.021 (0.005)
log capital	-0.019 (0.001)	0.026 (0.003)	-0.000 (0.000)	0.002 (0.001)	0.103 (0.002)	0.064 (0.008)	0.141 (0.017)	0.187 (0.113)	0.034 (0.001)	0.022 (0.005)
log employment	-0.003 (0.001)	0.002 (0.005)	0.001 (0.000)	0.002 (0.001)	0.129 (0.002)	0.110 (0.011)	-0.011 (0.018)	-0.399 (0.140)	0.034 (0.001)	0.008 (0.006)
log sales	-0.006 (0.001)	0.018 (0.004)	0.000 (0.000)	0.003 (0.001)	0.128 (0.002)	0.094 (0.010)	0.143 (0.018)	0.177 (0.137)	0.036 (0.001)	0.026 (0.005)
(c) Continuously innovative firms with more than 20 years of data										
log net worth	-0.038 (0.001)	0.043 (0.005)	-0.000 (0.000)	0.003 (0.001)	0.117 (0.003)	0.050 (0.011)	0.227 (0.027)	0.494 (0.179)	0.052 (0.001)	0.039 (0.009)
log capital	-0.051 (0.001)	0.050 (0.007)	-0.003 (0.000)	0.003 (0.001)	0.117 (0.003)	0.067 (0.014)	0.046 (0.026)	0.351 (0.230)	0.044 (0.001)	0.042 (0.010)
log employment	-0.044 (0.002)	-0.005 (0.010)	-0.003 (0.000)	0.004 (0.001)	0.153 (0.003)	0.161 (0.019)	-0.255 (0.030)	-0.874 (0.295)	0.050 (0.002)	0.013 (0.012)
log sales	-0.045 (0.001)	0.032 (0.008)	-0.003 (0.000)	0.005 (0.002)	0.138 (0.003)	0.130 (0.017)	-0.009 (0.027)	0.366 (0.283)	0.047 (0.001)	0.051 (0.011)

$o_{jt} = \alpha_j + \gamma \log s_{jt} + \epsilon_{jt}$. (c): Positive R&D over the last five years.

Pecking Order - Other FE

TABLE A.3
SOURCES OF VARIATION IN THE PECKING ORDER

	(1) Investment rate	(2) R&D share	(3) R&D -to-sales	(4) R&D activity	(5) Patent activity	(6) Patent-value -to-sales
(a) Baseline						
$\hat{\gamma}$	-0.068 (0.003)	0.024 (0.003)	0.002 (0.001)	0.016 (0.006)	0.049 (0.007)	0.021 (0.005)
N	45935	47286	41661	49105	49105	31176
R^2	0.263	0.857	0.877	0.854	0.639	0.678
(b) Sector fixed effects						
$\hat{\gamma}$	-0.014 (0.002)	0.019 (0.005)	0.006 (0.002)	0.073 (0.010)	0.155 (0.010)	0.062 (0.012)
N	45939	47290	41664	49109	49109	31177
R^2	0.084	0.586	0.494	0.549	0.360	0.261
(c) Sector-by-time fixed effects						
$\hat{\gamma}$	-0.0005 (0.003)	0.014 (0.005)	0.007 (0.002)	0.085 (0.011)	0.196 (0.011)	0.071 (0.014)
N	42962	44278	38785	46173	46173	28713
R^2	0.198	0.571	0.443	0.519	0.358	0.205
(d) Firm and time fixed effects						
$\hat{\gamma}$	-0.025 (0.003)	-0.007 (0.004)	0.003 (0.001)	0.018 (0.008)	0.097 (0.009)	0.021 (0.005)
N	45935	47286	41661	49105	49105	31176
R^2	0.318	0.862	0.878	0.855	0.643	0.691

(a): $\alpha_{jt} = \alpha_j + \gamma \log n_{jt} + \epsilon_{jt}$. (b): $\alpha_{jt} = \alpha_s + \gamma \log n_{jt} + \epsilon_{jt}$. (c): $\alpha_{jt} = \alpha_{st} + \gamma \log n_{jt} + \epsilon_{jt}$. (d): $\alpha_{jt} = \alpha_j + \alpha_t + \gamma \log n_{jt} + \epsilon_{jt}$.

Sample Lifecycle of a Firm in the Model

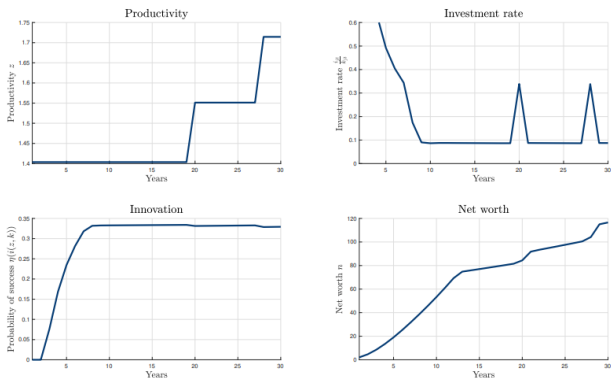


Figure: $\epsilon_{jt} = 0$ for all firms. In initial years has high investment rate and does not innovate. As ages, exhausts MPK and increases innovation. Firm receives two successful innovations, both accompanied by investment spikes.

Not De-Trended FOCs

$$1 + \lambda_t(z, n) = \frac{1}{1 + r_t} \mathbb{E}_t \left[(MPK_{t+1}(z', k') + 1 - \delta) \times (1 + (1 - \pi_d) \lambda_{t+1}(z', n')) \right] + \theta \mu_t(z, n)$$

$$1 + \lambda_t(z, n) \geq \frac{\eta'(i)}{(A_t z)^{\frac{1}{1-\alpha}}} \frac{1}{1 + r_t} (\mathbb{E}_t [v_{t+1}(z', n' \mid \iota_{t+1}(z, n) = 1] - \mathbb{E}_t [v_{t+1}(z', n' \mid \iota_{t+1}(z, n) = 0)])$$

with = if $i_t(z, n) > 0$

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Decomposing the Return to Capital

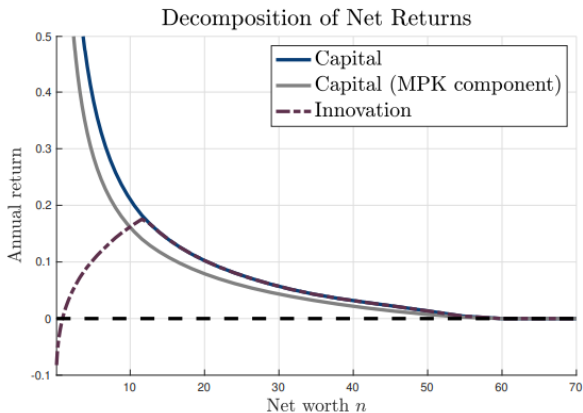


Figure: Net Return = RHS of Euler equation - 1

Firm's Decision Rules - Bellman

- Detrend using $Z_t = (A_t \int z_{jt} dj)^{\frac{1}{1-\alpha}} = (\int z_{jt} dj)^{\frac{1+\alpha}{1-\alpha}}$. Divide the Bellman by Z_t

$$\begin{aligned} \frac{v_t^{\text{cont}}(z, n)}{Z_t} = \max_{k', i, b'} \quad & \frac{n}{Z_t} - \frac{k'}{Z_t} - \frac{(A_t z)^{\frac{1}{1-\alpha}} i}{Z_t} + \frac{b'}{Z_t (1+r_t)} \\ & + \frac{1}{1+r_t} \mathbb{E}_t \left[\pi_d \frac{n'}{Z_t} + (1-\pi_d) \frac{v_{t+1}^{\text{cont}}(z', n')}{Z_t} \right] \end{aligned}$$

and write in terms of detrended variables and growth rate $g_t = \frac{Z_{t+1}}{Z_t}$ only:

$$\begin{aligned} \frac{v_t^{\text{cont}}(z, n)}{Z_t} = \max_{k', i, b'} \quad & \tilde{n} - (1+g_t) \tilde{k}' - \tilde{z}^{\frac{1}{1-\alpha}} i + \frac{(1+g_t) \tilde{b}'}{(1+r_t)} \\ & + \frac{1+g_t}{1+r_t} \mathbb{E}_t \left[\pi_d \tilde{n}' + (1-\pi_d) \frac{v_{t+1}^{\text{cont}}(z', n')}{Z_{t+1}} \right]. \end{aligned}$$

Firm's Decision Rules - Bellman

- Law of motion for z :

$$\log \frac{z}{\int z_{jt+1} dj} = \log \frac{z}{\int z_{jt+1} dj} + \Delta + \varepsilon_{jt+1} \Rightarrow \log \tilde{z}' = \log \frac{\tilde{z}}{1 + \tilde{g}_t} + \Delta + \varepsilon_{jt+1}$$

where $\tilde{g}_t = \frac{\int z_{jt+1} dj}{\int z_{jt} dj}$ is growth rate of firm-specific productivity.

- Lagrangian of detrended Bellman:

$$\begin{aligned} \mathcal{L} = & (1 + \lambda_t(\tilde{z}, \tilde{n})) \left(\tilde{n} - (1 + g_t) \tilde{k}' - \tilde{z}^{\frac{1}{1-\alpha}} i + \frac{(1 + g_t) \tilde{b}'}{(1 + r_t)} \right) \\ & + (1 + g_t) \mu_t(\tilde{z}, \tilde{n}) \left(\theta \tilde{k}' - \tilde{b}' \right) \\ & + \chi_t(\tilde{z}, \tilde{n}) i + \frac{1 + g_t}{1 + r_t} \mathbb{E}_t \left[\pi_d \tilde{n}' + (1 - \pi_d) \tilde{v}_{t+1} (\tilde{z}', \tilde{n}') \right] \end{aligned}$$

Firm's Decision Rules - FOCs

- FOC for borrowing $\tilde{b}'$:

$$(1 + \lambda_t(\tilde{z}, \tilde{n})) \frac{1 + g_t}{1 + r_t} = (1 + g_t) \mu_t(\tilde{z}, \tilde{n}) - \frac{1 + g_t}{1 + r_t} \mathbb{E}_t \left[\pi_d \frac{\partial \tilde{n}'}{\partial \tilde{b}'} + (1 - \pi_d) \frac{\partial \tilde{v}_{t+1}(\tilde{z}', \tilde{n}')}{\partial \tilde{n}'} \frac{\partial \tilde{n}'}{\partial \tilde{b}'} \right]$$

Since $\frac{\partial \tilde{v}_t(\tilde{z}, \tilde{n})}{\partial \tilde{n}'} = 1 + \lambda_t(\tilde{z}, \tilde{n})$ and $\frac{\partial \tilde{n}'}{\partial \tilde{b}'} = -1$

$$(1 + \lambda_t(\tilde{z}, \tilde{n})) \frac{1 + g_t}{1 + r_t} = (1 + g_t) \mu_t(\tilde{z}, \tilde{n}) + \frac{1 + g_t}{1 + r_t} \mathbb{E}_t [\pi_d + (1 - \pi_d) (1 + \lambda_{t+1}(\tilde{z}', \tilde{n}'))]$$

Simplify:

$$\lambda_t(\tilde{z}, \tilde{n}) = (1 + r_t) \mu_t(\tilde{z}, \tilde{n}) + (1 - \pi_d) \mathbb{E}_t \lambda_{t+1}(\tilde{z}', \tilde{n}')$$

Firm's Decision Rules - FOCs

- FOC for $\tilde{k}'$ is

$$(1 + g_t)(1 + \lambda_t(\tilde{z}, \tilde{n})) = \theta(1 + g_t)\mu_t(\tilde{z}, \tilde{n}) + \frac{1 + g_t}{1 + r_t} \mathbb{E}_t \left[\pi_d \frac{\partial \tilde{n}'}{\partial \tilde{k}'} + (1 - \pi_d) \frac{\partial \tilde{v}_{t+1}(\tilde{z}', \tilde{n}')}{\partial \tilde{n}'} \frac{\partial \tilde{n}'}{\partial \tilde{k}'} \right]$$

Since $\frac{\partial \tilde{n}'}{\partial \tilde{k}'} = MPK(\tilde{z}', \tilde{k}') + (1 - \delta)$, where $MPK(\tilde{z}', \tilde{k}') = \alpha \tilde{z}' (\tilde{k}')^{\alpha-1}$

$$1 + \lambda_t(\tilde{z}, \tilde{n}) = \theta \mu_t(\tilde{z}, \tilde{n}) + \frac{1}{1 + r_t} \mathbb{E}_t \left[\left(MPK(\tilde{z}', \tilde{k}') + (1 - \delta) \right) (1 + (1 - \pi_d) \lambda_{t+1}(\tilde{z}', \tilde{n}')) \right]$$

- FOC for i is

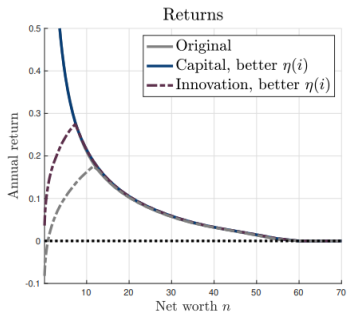
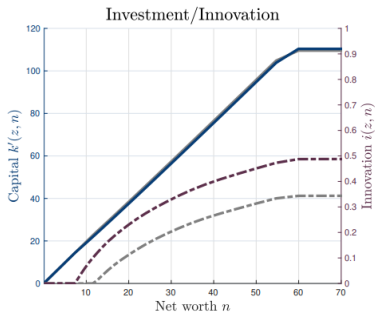
$$(1 + \lambda_t(\tilde{z}, \tilde{n})) \tilde{z}^{\frac{1}{1-\alpha}} = \chi_t(\tilde{z}, \tilde{n}) + \frac{1 + g_t}{1 + r_t} \frac{\partial}{\partial i} \mathbb{E}_t [\pi_d \tilde{n}' + (1 - \pi_d) \tilde{v}_{t+1}(\tilde{z}, \tilde{n}')]]$$

$$(1 + \lambda_t(\tilde{z}, \tilde{n})) \tilde{z}^{\frac{1}{1-\alpha}} \geq \frac{1 + g_t}{1 + r_t} \eta'(i) \left[\frac{\pi_d (E^\epsilon [\tilde{n}' | \iota = 1] - E^\epsilon [\tilde{n}' | \iota = 0]) + (1 - \pi_d) (E^\epsilon [\tilde{v}_{t+1}(\tilde{z}', \tilde{n}') | \iota = 1] - E^\epsilon [\tilde{v}_{t+1}(\tilde{z}', \tilde{n}') | \iota = 0])}{(1 - \pi_d) (E^\epsilon [\tilde{v}_{t+1}(\tilde{z}', \tilde{n}') | \iota = 1] - E^\epsilon [\tilde{v}_{t+1}(\tilde{z}', \tilde{n}') | \iota = 0])} \right],$$

with equality if $i > 0$.

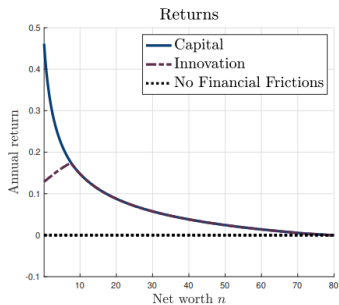
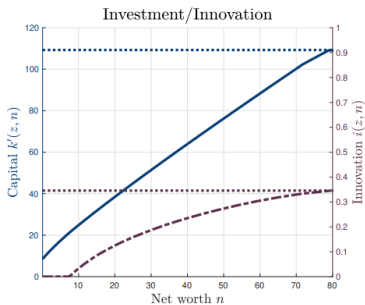
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Higher Success Probability $\eta(i)$ given i



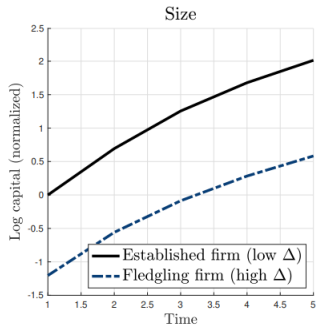
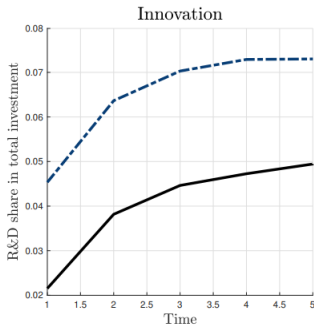
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Collateralizable Ideas



$$b_{jt+1} \leq \tilde{\theta} \mathbb{E}_t \left[A_{t+1} z_{jt+1} k_{jt+1}^\alpha \right]$$

Heterogeneity in Innovation Size



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Decision Rules for Different z

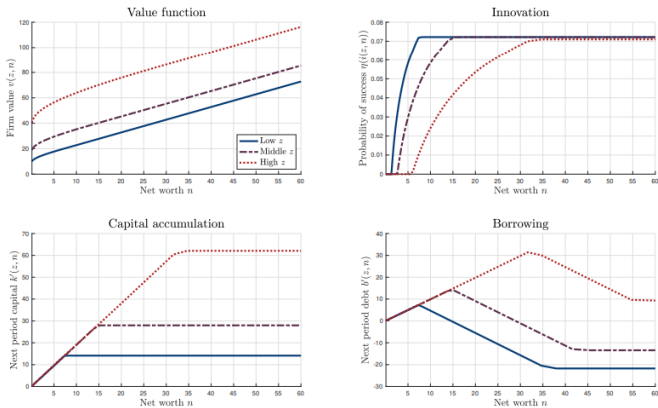


Figure: Firm decision rules in the market BGP. All variables have been detrended.

Calibration: Fixed Parameters

FIXED PARAMETERS		
Parameter	Description	Value
<i>Household</i>		
β	Discount factor	0.97
$1/\sigma$	EIS	1.50
<i>Firms</i>		
α	Output elasticity w.r.t inputs	0.55
δ	Depreciation rate	0.08
π_d	Exit rate	0.08

Figure: EIS high: To reduce interest-sensitivity of investment.

π_d matches exit rates in Business Dynamics Statistics (BDS) and Compustat sample.

» Calibration

Optimal Allocation of Capital

- Idea: Hold aggregate capital fixed, but distributes it across firms to maximize current output given the current distribution of productivity.

$$\frac{Y_t}{Y_t^*} = \frac{\int z_{jt} \left(\frac{k_{jt}}{K_t} \right)^\alpha dj}{\left(\int z_{jt}^{\frac{1}{1-\alpha}} dj \right)^{1-\alpha}}$$

- FOC with respect to k_{jt} :

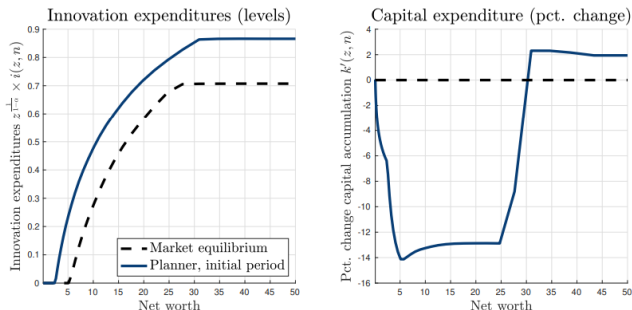
$$k_{jt} = \left(\frac{\alpha A_t z_{jt}}{\lambda} \right)^{\frac{1}{1-\alpha}}$$

- Integrating across firms j and using $\int k_{jt} dj = K_t$ gives λ , which plugged into FOC gives:

$$\frac{k_{jt}}{K_t} = \frac{z_{jt}^{\frac{1}{1-\alpha}}}{\int z_{jt}^{\frac{1}{1-\alpha}} dj}$$

$$TFP_t = A_t \int z_{jt} \left(\frac{k_{jt}}{K_t} \right)^\alpha dj$$

Firm-Level Allocations Chosen by Planner



Notes: decision rules in market equilibrium vs. constrained-efficient allocation in initial period of the transition, for a given level of productivity z . Left panel plots innovation expenditures $(A_t z)^{\frac{1}{1-\alpha}} i(z, n)$ for a firm with average productivity z for a given level of net worth. Dashed black line is the private policy rule in the market equilibrium and solid blue line is the planner's policy rule. Right panel plots the percentage difference between the planner's capital accumulation policy relative to the private policy in the market equilibrium.